
Network Hardening Best Practices (Reference for RAG)

1. Segmentation and Least Privilege

- Separate user, management, and service planes.
- Permit only required east-west and north-south flows.
- Apply deny-by-default in forwarding policy.

2. Router/Switch Control Plane Protection

- Disable unnecessary services (for example telnet).
- Restrict management protocols to trusted sources.
- Drop invalid conntrack state packets.

3. Baseline Kernel/Network Stack Hardening

- Disable ICMP redirects handling where unnecessary.
- Disable sending redirects from routers.
- Ignore broadcast ICMP echo requests.

4. Monitoring and Verification

- Verify policy outcomes with active probes (ping/tcp tests).
- Collect change records and implementation timestamps.
- Confirm both intended connectivity and intended blocks.

5. Secure Change Management

- Make changes idempotent (safe to re-apply).
- Store exact command/action logs for each request.
- Provide explicit success/failure confirmation.

6. Common Immediate Controls

- Block plaintext protocols (telnet, legacy cleartext mgmt).
- Limit ICMP where abuse is a concern, preserving diagnostics.
- Enforce source/destination based forwarding filters.

7. Operational Notes

- Test in emulated topologies before production rollout.
- Keep an emergency rollback path for policy changes.
- Reassess hardening controls after topology changes.